



Questions

#0: Start a Spark Session and import the libraries you will need.

Read the stock data from the GCP bucket and save it as a data frame.

#1: Display the Schema of the data frame.

#2: Display the summary statistics for all columns in the data frame.

#3: Find and display the five records that has the highest price in the High column.

#4: What day had the lowest price? Display the date (as given in the Date column) and the price?

#5: What is the range of the Volume column? (Hint: Range is the difference between max and min values)

#6: Calculate the average of the High column. What percent of the observations had a High price greater than the average?

#7: What is the total Volume per year? Sort the results based on the total Volume in a decreasing order.

#8: What is the average Adjusted Close price for each month? Sort the results in an increasing order based on the average Adjusted Close price.

#9: Using SQL commands, create a new data frame that includes the records for the days where closing price is lower than the opening price. Display the number of records in this new data frame.

#10: Perform the following using SQL commands: Count the number of days in each month of each year during the last three years, where closing price is at least \$5 less than the opening price. The resulting data frame should be sorted based on Year first and then Month in increasing order.